



Investigation of Primitive-Space Scene Change Detection in Under- water 3D Gaussian Splatting Models

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Understanding the Concept

3D Gaussian Splatting for Real-Time Radiance Field Rendering

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Fig. 1. Our method achieves real-time rendering of radiance fields with quality that equals the previous method with the best quality [Barron et al. 2022], while only requiring optimization times competitive with the fastest previous methods [Fridovich-Keil and Yu et al. 2022; Müller et al. 2022]. Key to this performance is a novel 3D Gaussian scene representation coupled with a real-time differentiable renderer, which offers significant speedup to both scene optimization and novel view synthesis. Note that for comparable training times to InstantNGP [Müller et al. 2022], we achieve similar quality to theirs; while this is the maximum quality they reach, by training for 51min we achieve state-of-the-art quality, even slightly better than Mip-NeRF360 [Barron et al. 2022].

Radiance Field methods have recently revolutionized novel-view synthesis of scenes captured with multiple photos or videos. However, achieving high visual quality still requires neural networks that are costly to train and render, while recent faster methods inevitably trade off speed for quality. For unbounded and complete scenes (rather than isolated objects) and 1080p resolution rendering, no current method can achieve real-time display rates. We introduce three key elements that allow us to achieve state-of-the-art visual quality while maintaining competitive training times and importantly allow high-quality real-time (≥ 30 fps) novel-view synthesis at 1080p resolution. First, starting from sparse points produced during camera calibration, we represent the scene with 3D Gaussians that preserve desirable properties of continuous volumetric radiance fields for scene optimization while avoiding unnecessary computation in empty space; Second, we perform interleaved optimization/density control of the 3D Gaussians, notably optimizing anisotropic covariance to achieve an accurate representation of the scene; Third, we develop a fast visibility-aware rendering algorithm that supports anisotropic splatting and both accelerates training and allows real-time rendering. We demonstrate state-of-the-art visual quality and real-time rendering on several established datasets.

CCS Concepts: • Computing methodologies → Rendering, Point-based models; Rasterization; Machine learning approaches.

*Both authors contributed equally to the paper.

Additional Key Words and Phrases: novel view synthesis, radiance fields, 3D gaussians, real-time rendering

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1 INTRODUCTION

Meshes and points are the most common 3D scene representations because they are explicit and are a good fit for fast GPU/CUDA-based rasterization. In contrast, recent Neural Radiance Field (NeRF) methods build on continuous scene representations, typically optimizing a Multi-Layer Perceptron (MLP) using volumetric ray-marching for novel-view synthesis of captured scenes. Similarly, the most efficient radiance field solutions to date build on continuous representations by interpolating values stored in, e.g., voxel [Fridovich-Keil and Yu et al. 2022] or hash [Müller et al. 2022] grids or points [Xu et al. 2022]. While the continuous nature of these methods helps optimization, the stochastic sampling required for rendering is costly and can result in noise. We introduce a new approach that combines the best



Understanding the Concept

From Pixels to Primitives: Scene Change Detection in 3D Gaussian Splatting

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Abstract

Scene change detection methods built on Gaussian splatting universally follow a render-then-compare paradigm: the pre-change scene is rendered into 2D and compared against post-change images via pixel or feature residuals. This change detection problem with Gaussian Splatting has been treated as a question about pixels: we treat it as a question about primitives. We provide direct evidence that native primitive attributes alone – position, anisotropic covariance, and color – carry sufficient signal for scene change detection. What makes primitive-space comparison hard is the under-constrained nature of Gaussian splatting representation: independent optimizations yield primitive solutions whose count, positions, shapes, and colors differ even where nothing has changed. We address this challenge with anisotropic models of geometric and photometric drift, complemented by a per-primitive observability term that reflects the extent to which each Gaussian is constrained by the camera geometry. Operating directly on primitives gives our method, GS-Diff, two properties that distinguish it from render-then-compare methods. First, change maps are multi-view consistent by construction, where prior work had to learn this through an additional optimization objective. Second, geometric and appearance changes are scored separately, identifying not just *where* but *what kind* of change occurred, distinguishing structural changes (e.g., an added object) from surface-level ones (e.g., a color change) without supervision or external model dependencies. On real-world benchmarks, GS-Diff surpasses the prior state-of-the-art approach by $\sim 17\%$ in mean Intersection over Union. Code and annotations will be released upon acceptance.

1 Introduction

3D Gaussian splatting (3DGS) [28] has become a foundational representation for photorealistic digital twins [17], with adoption spanning spatial maps [30], heritage preservation [8], and industrial asset management [13]. A critical challenge in these applications is understanding how environments evolve over time: an agent visits a scene at different times, capturing images from unconstrained viewpoints, and must detect and localize changes that have occurred between inspections. This *Multi-View Scene Change Detection* (SCD) problem [13,13,35] requires robustness to arbitrary camera trajectories and the ability to distinguish genuine structural or appearance changes from irrelevant variations such as lighting shifts, shadows, and reflections.

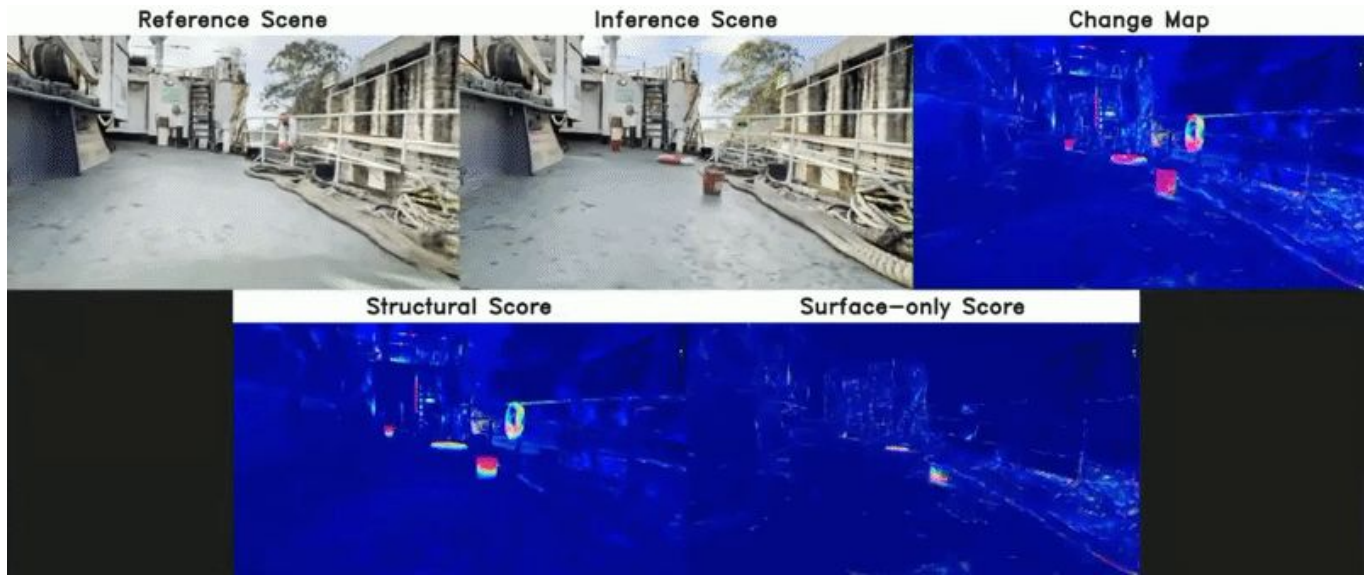
Existing work [13,35,25,30,33,11,15] follows a *render-then-compare* paradigm, treating 3DGS as a *rendering engine*: the pre-change scene is reconstructed, rendered at post-change poses, and compared to post-change images via foundation-model features [37,39], pixel-level metrics [47], or combinations thereof (Figure 1). Because genuine changes register consistently across views while distractors such as shadows and reflections fire inconsistently, state-of-the-art methods aggregate change cues across views to enforce multi-view consistency [13,13,35,49,11], suppressing these

Note

Structural = Geometric Kernel = Physical Change

Surface = Appearance Kernel = Colour Change

Final Output



Extending the Concept

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SeaSplat: Representing Underwater Scenes with 3D Gaussian Splatting and a Physically Grounded Image Formation Model

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Abstract—We introduce SeaSplat, a method to enable real-time rendering of underwater scenes leveraging recent advances in 3D radiance fields. Underwater scenes are challenging visual environments, as rendering through a medium such as water introduces both range and color dependent effects on image capture. We constrain 3D Gaussian Splatting (3DGS), a recent advance in radiance fields enabling rapid training and real-time rendering of full 3D scenes, with a physically grounded underwater image formation model. Applying SeaSplat to the real-world scenes from SeaThru-NcRF dataset, a scene collected by an underwater vehicle in the US Virgin Islands, and simulation-degraded real-world scenes, not only do we see increased quantitative performance on rendering novel viewpoints from the scene with the medium present, but are also able to recover the underlying true color of the scene and restore renders to be without the presence of the intervening medium. We show that the underwater image formation helps learn scene structure, with better depth maps, as well as show that our improvements maintain the significant computational improvements afforded by leveraging a 3D Gaussian representation. Code, data, and visualizations are available at <https://seasplat.github.io>

I. INTRODUCTION

Underwater vehicles equipped with imaging capabilities have increasingly been deployed for complex and adaptive tasks including environmental monitoring [1]–[3], visual target tracking [4], [5], and exploring underwater environments [6], [7]. However, visual perception underwater presents a significantly different set of characteristics than images collected in air, making it challenging to directly utilize many advances in computer vision to this domain [8], [9].

Images collected underwater exhibit degradation because of the inherent effects of imaging through a medium, water. Light propagating through water is affected by range-dependent and spectrally-sensitive attenuation as well as backscatter. In practice, the attenuation causes underwater images to be lacking in certain colors, specifically red, while having an excess of others, blue and green, while the backscatter causes underwater images to exhibit a veiling or

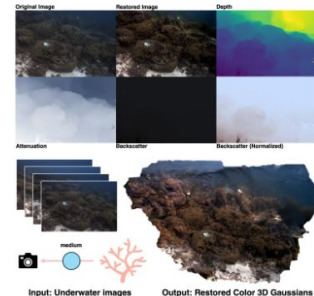
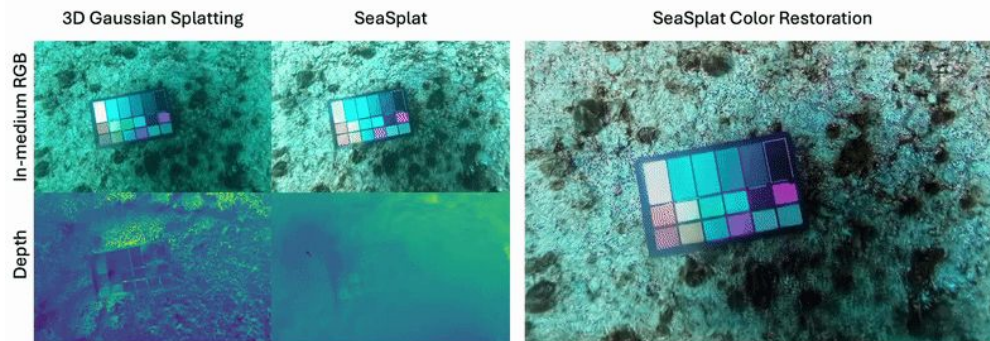


Fig. 1: Overview SeaSplat constrains a 3D Gaussian representation with a physics-based underwater image formation model to restore the scene against effects caused by imaging through water and provide higher quality novel view synthesis. SeaSplat estimates medium parameters for the depth and wavelength dependent effects of backscatter and attenuation, while also enabling rapid view synthesis of the scene with and without the medium.

More recently, 3D Gaussian Splatting [11] has seen success in enabling real-time, photorealistic rendering of these 3D environments. However, these methods typically assume atmospheric (e.g., imaging through air) conditions for the data where color is relatively constant with distance, leading to undesirable effects such as floaters and other geometric inconsistencies when applied to underwater environments. We propose SeaSplat, a method that combines 3D Gaussian Splatting with a physically grounded underwater image formation model. By simultaneously learning the parameters of the medium along with the underlying 3D representation,

Alternative methods such as Watersplatting(2025) exist



Water noise

Water can be subtracted from output as well

Extending the Concept

SuGaR: Surface-Aligned Gaussian Splatting for Efficient 3D Mesh Reconstruction and High-Quality Mesh Rendering

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Figure 1. We introduce a method that extracts accurate and editable meshes from 3D Gaussian Splatting representations within minutes on a single GPU. The meshes can be edited, animated, composited, etc. with very realistic Gaussian Splatting rendering, offering new possibilities for Computer Graphics. Note for example that we changed the posture of the robot between the captured scene on the bottom left and the composited scene on the right. The supplementary material provides more examples, including a video illustrating our results.

Abstract

We propose a method to allow precise and extremely fast mesh extraction from 3D Gaussian Splatting [15]. Gaussian Splatting has recently become very popular as it yields realistic rendering while being significantly faster to train than NeRFs. It is however challenging to extract a mesh from the millions of tiny 3D Gaussians as these Gaussians tend to be unorganized after optimization and no method has been proposed so far. Our first key contribution is a regularization term that encourages the Gaussians to align well with the surface of the scene. We then introduce a method that exploits this alignment to extract a mesh from the Gaussians using Poisson reconstruction, which is fast, scalable, and preserves details, in contrast to the Marching Cubes algorithm usually applied to extract meshes from Neural SDFs. Finally, we introduce an optional refinement strategy that binds Gaussians to the surface of the mesh, and jointly optimizes these Gaussians and the mesh through Gaussian splatting rendering. This enables easy editing, sculpting, animating, and relighting of the Gaussians by manipulating the mesh instead of the Gaussians themselves. Retrieving such an editable mesh for realistic rendering is done within minutes with our method, compared to hours with the state-of-the-art method on SDFs, while providing a better rendering quality.

Erratum

We identified a minor typographical error in Subsection 4.1 in the earlier version of the paper. In the computation of our regularization term $\mathcal{R}$ in Equation 8, we use $p \rightarrow \pm s_{gs} \sqrt{-2 \log(d(p))}$ instead of $p \rightarrow \pm s_{gs} \sqrt{-2 \log(d(p))}$ as an ‘ideal’ distance function associated with the density d (Equation 7). As detailed in the paper, this distance function aligns with the true surface of the scene in an ideal scenario where $d = \bar{d}$. We have updated Equation 7 to clarify this matter.

1. Introduction

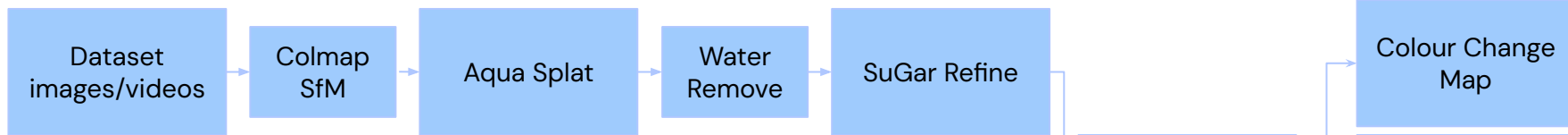
After NeRFs [22], 3D Gaussian Splatting [15] has recently become very popular for capturing a 3D scene and rendering it from novel points of view. 3D Gaussian Splatting optimizes the positions, orientations, appearances (represented as spherical harmonics), and alpha blending of many tiny 3D Gaussians on the basis of a set of training images of the scene to capture the scene geometry and appearance. Because rendering the Gaussians is much faster than rendering a neural field, 3D Gaussian Splatting is much faster than NeRFs and can capture a scene in a few minutes.

While the Gaussians allow very realistic renderings of the scene, it is still however challenging to extract the sur-



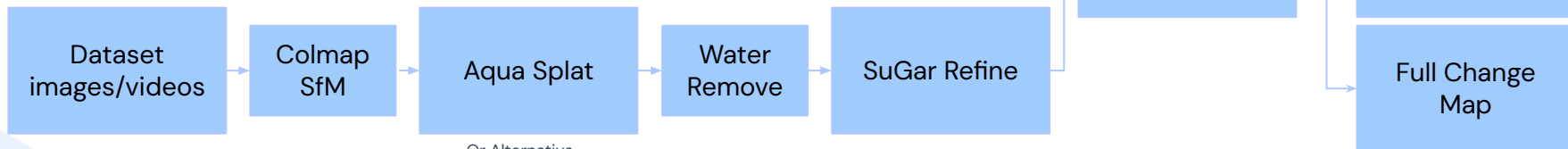
Proposed Pipeline

Collection time A



Or Alternative

Collection time B



Or Alternative

ExoRRAP Dataset
here

Above:

Proposed custom underwater pipeline for GS-DIFF application which will be in development till end of 2027

Why the EcoRRAP dataset

Data Type

- 4D photogrammetry based capture has great SfM foundation which is required for our method
- Ideally no moving features in the dataset if captured for SfM
- Great coverage of features for Gaussian training
- Identical locations for perfect Change detection dataset
- Method works even if capture path is not identical, because it is we will be able to evaluate the method in great detail

Application

- Researchers can view a heat map of change and investigate in more detail via the 4D models
- Colour and geometric separation can occur so focus on bleaching or growth can be separate
- Real time rendering so the dataset can be walked through in demonstrations of your work from a simple browser
- The gaussian model is lightweight and can be shared quick and effectively

Potential Blockers

- To much change between timestamps and the whole output is bright and unusable
- Moving features in the dataset like sea animals or sand, Moved items duplicating on change map cluttering the output
- If water or geometric noise cannot be removed completely errors will propagate into the GS-DIFF model
- Coral is too geometrically complex to be modeled via gaussian noise while being measured under noise